

UGTS-KC 3.9.3

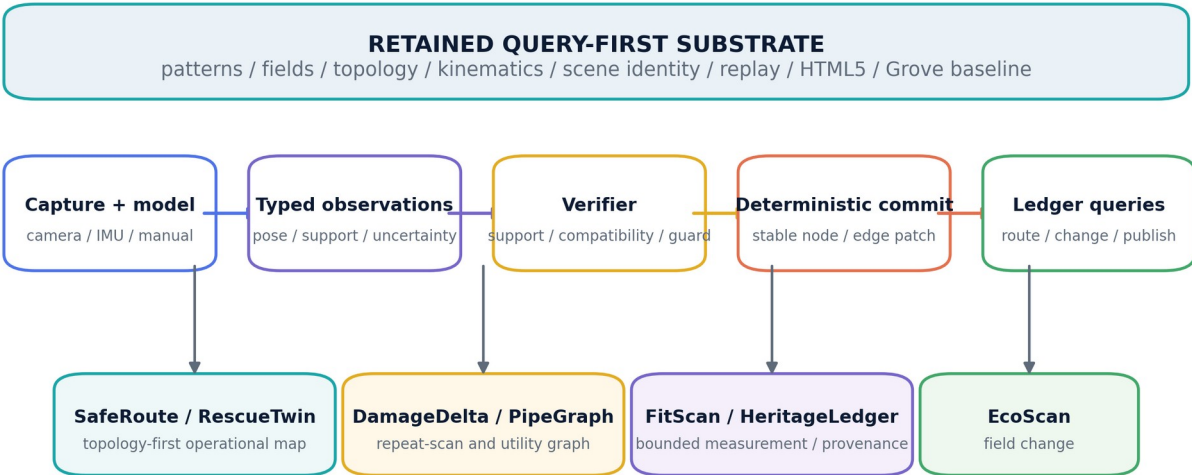
KC ATLAS

Pocket Spatial Ledger

Spatial Evidence, Accessibility, Repeat-Scan Change
and High-Impact Application Profiles

UGTS-KC 3.9.3 KC Atlas architecture

A phone or model proposes evidence. The ledger alone commits authoritative spatial change.



Canonical authority: support -> compatibility -> confidence/error guard -> verified event -> transition -> lineage

Release field	Value
Prepared for	Tom Klootwijk
Release	Version 3.9.3 Codename KC Atlas
Ledger schema	ugts-kc-spatial-ledger-3.9.3
Mechanism range	M001-M589 by combined source/catalog
Android decision	Retain 3.9.2 Grove baseline; postpone new capture app

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Earlier requester-supplied identity and substrate-attribution records remain preserved in the retained package. This report is a technical design and executable reference artifact, not legal proof of identity, ownership, patentability, clinical fitness, structural safety or physical-device performance.

Contents and Executive Summary

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Executive summary

KC Atlas turns the project's query-first geometry, topology, scene identity, deterministic commit, replay and offline-export mechanisms into a common spatial evidence substrate. The principal engineering result is not another mesh format. It is a ledger that can state what was observed, where it was supported, how uncertain it was, which event changed the authoritative map and how the resulting state can be replayed or compared.

RELEASE DECISION

GO for substrate expansion first. Version 3.9.3 delivers the shared evidence, route, change and publication layer. The new Android camera/IMU/model application is postponed until this contract is frozen; the attached working Grove implementation is retained unchanged as its native baseline.

Measured expansion

Metric	3.9.3 result
Mechanisms	60 new entries M530-M589; combined catalog reaches M589.
Tests	71 new tests; 347/347 retained-plus-new tests pass.
Runtime	5 new Python modules, 3,298 lines; complete Python source is 14,408 lines.
Profiles	7 content-addressed application profiles.
Demonstration	3 accepted events; 16 change candidates; replay hash equals current state hash.
Exports	Self-contained HTML + canonical JSON + SHA-256 manifest.
Distributions	Wheel and source archive built and clean-install tested.

1. Release Definition and Decision

Version 3.9.3 is additive over the supplied K-Kij-T / Grove 3.9.2 package. It retains the 2D vector game stack, mobile-3D records, scene and geometry compiler, spatial indexing, two-hand interaction, deterministic runtime/replay, browser export and native Android Grove project. KC Atlas adds a separate evidence-bearing spatial-map path rather than forcing capture semantics into the game-project schema.

```
capture profile + typed observations
-> finite spatial support
-> semantic / route / policy compatibility
-> confidence + numerical-error guard
-> verified proposal
-> deterministic event commit
-> stable node / edge / measurement patch
-> lineage + checkpoint + replay
-> route, change and publication queries
```

1.1 What changed

- A versioned spatial-ledger record with captures, observations, stable nodes, route edges, events and checkpoints.
- Closed scalar measurement intervals and pose uncertainty instead of unsupported point precision.
- Finite AABB, sphere, cone and frustum support volumes evaluated before semantic compatibility and event guards.
- Deterministic map proposals, explicit rejection reasons, same-time conflict policy and hash-verified replay.
- Constrained route and reachability queries with passable/blocked/restricted/unknown semantics.
- Repeat-scan change reports keyed by stable identity and uncertainty intervals.
- Application profile readiness, publication redaction and a no-network evidence bundle.
- A future Android/vision adapter packet that is proposal-only and cannot mutate the ledger directly.

1.2 Compatibility guarantee

Area	Retained 3.9.2 behavior	3.9.3 addition
Identity	Scene node, asset and lineage identity remain distinct.	Stable map-node, route-edge, capture and observation IDs.
Runtime	Fixed-step game/runtime proposal and replay APIs remain available.	Spatial proposals are exported as SpatialEventProposal to avoid API shadowing.
Delivery	HTML5, glTF, scene pack and Android Grove paths remain.	Offline spatial evidence bundle and publication view.
Distribution	Python distribution name retained for install compatibility.	Runtime edition and schemas identify KC Atlas 3.9.3.
Android	Grove project and supplied APK preserved.	New capture app explicitly postponed.

2. Source Continuity and Evidence Discipline

The project sources converge on a stable authority rule: geometry and topology generate typed candidates; local support removes irrelevant candidates; compatibility decides whether states may couple; a finite guard and numerical policy classify an event; deterministic transition logic updates state and lineage. KC Atlas preserves that rule and uses it to protect the map from unchecked perception output.

Registered source	3.9.3 use	SHA-256 prefix
3.9 KC Elizabeth Vector Game Runtime	Retained project model, fixed-step runtime, self-contained HTML5 export and validation discipline.	d7ee3c91abf60e35...
3.6.2 SCLP	Finite cone/sphere supports, log-polar calculus, bounded jitter, 64-bit layouts and prefix refinement.	6987490a0a7b7ed7...
2.0 Expanded Substrate	Pattern/field calculus, route identity, SE(3), bounded dynamics, event degeneracy and error contracts.	a29dd9959ebd300c...
Two Hands 3.0	Stable scene/node identity, geometry compiler, BVH, independent error budgets, proposal commit, replay and interchange.	183ffa4e2bf29cfa...
1.1 GPU-native Addendum	Canonical query-first sequence, packed ABI, measured-performance boundaries and narrow one-bit semantics.	3a6b6c57118ca4e2...
3.6 Literal Referential Substrate	Content-addressed definitions, explicit dependency graph, operation order and provenance.	e21433cd8378368d...
3.9.2 K-Kij-T / Grove complete files	Immediate implementation baseline, retained Android source and supplied APK.	2d0197bb4526ca99...

2.1 Source-derived continuity

- UGTS-GN 1.1 fixes the query-first order, narrow one-bit semantics, precision rule and physical-performance evidence boundary.
- UGTS-KC 2.0 supplies pattern/field calculus, route-class identity, kinematics, bounded dynamics and explicit event degeneracy handling.
- KC Two Hands 3.0 supplies stable scene identity, AABB/BVH indexing, independent geometry error budgets, proposal/commit, checkpoints, replay and interchange.
- UGTS-KC 3.6 makes definitions, dependencies, provenance and operation order first-class substrate objects.
- SCLP 3.6.2 supplies finite log-polar calculus, interval guards, exact 64-bit layouts and prefix refinement with explicit limits.
- KC Elizabeth 3.9 demonstrates one project feeding deterministic simulation and a self-contained, offline HTML5 result.

2.2 Engineering delta

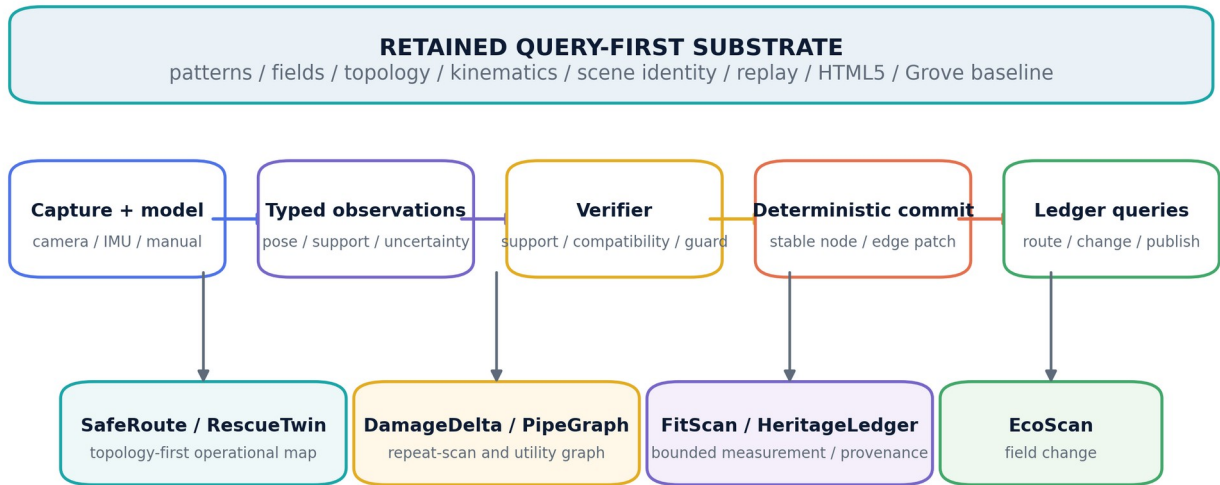
EVIDENCE CLASSIFICATION

The seven application profiles, spatial-ledger implementation and future perception packet are 3.9.3 engineering additions inferred from the retained mechanisms. The source PDFs do not claim that these products, a monocular SLAM model or a phone capture application were already completed.

3. Formal KC Atlas Architecture

UGTS-KC 3.9.3 KC Atlas architecture

A phone or model proposes evidence. The ledger alone commits authoritative spatial change.



Canonical authority: support -> compatibility -> confidence/error guard -> verified event -> transition -> lineage

Figure 1. The retained query-first substrate remains authoritative. Capture and perception produce typed proposals; the ledger commits evidence-bearing spatial change.

UGTS-KC 3.9.3 = UGTS-KC 3.9.2 + (L, U, Q, Delta, A, X, B)

L spatial evidence ledger
 U interval and pose uncertainty
 Q route / reachability queries
 Delta repeat-scan change calculus
 A application-profile registry
 X offline publication / evidence exchange
 B future Android perception bridge boundary

3.1 Authoritative ledger

The minimum authoritative object is:

L = (metadata, captures, observations, nodes, edges, events, checkpoints)

Raw video, meshes, depth maps, segmentation tensors and display graphics remain optional referenced evidence or downstream views. A ledger may be useful before a complete mesh exists because route topology, door state, clearance intervals and unknown regions are first-class records.

4. Spatial Ledger Record Model

Record	Required authority role	Key guards
LedgerMetadata	Stable ledger ID, title, author, coordinate system, units, application profiles and governance metadata.	Schema/version, finite units, canonical JSON.
CaptureProfile	Device/model/calibration/privacy context for one acquisition session.	Positive units, metric anchor, model/calibration hash, privacy consistency.
Observation	Evidence proposal with semantic class, frames, pose, support, confidence, error and provenance.	Existing capture, finite numbers, hash verification.
MapNode	Stable room/door/component/landmark identity with measurements and lineage.	Unique ID, typed pose, interval-valid measurements, evidence references.
RouteEdge	Stable topological connection with status, modes, length, clearance, slope and evidence.	Existing endpoints, non-self-loop, passability semantics.
LedgerEvent	Accepted or rejected ordered proposal with patch and pre/post hashes.	Monotonic sequence, deterministic ordering, conflict reason.
Checkpoint	Snapshot hash at an event sequence.	Snapshot/state hash equality.
ChangeReport	Stable-ID comparison and uncertainty-aware classification.	Baseline/current hashes and canonical candidate ordering.

4.1 Identity split

```

node identity      != coordinate
route-edge identity != current passability
observation identity != semantic target identity
render mesh       != authoritative measurement
content hash      != lineage or legal ownership

```

This split follows the retained scene/asset/lineage discipline. A door may be remeasured or change state without becoming a new logical door. Two routes may share coordinates while remaining different route classes or floors.

4.2 Canonical state identity

The ledger state hash covers metadata, captures, observations, nodes and edges in canonical sorted JSON. Event history is intentionally excluded from the state identity, allowing a directly constructed state to be compared with replay. Events and checkpoints remain separately hash-checked evidence of how that state was reached.

5. Capture and Perception Adapter Boundary

5.1 Capture profile

Field	Purpose
device_model / sensor_mode	Declare the acquisition endpoint and mode; never infer accuracy from a product name.
coordinate_system / units_per_meter	Prevent silent handedness, axis and scale mismatches.
calibration_hash / model_hash	Bind observations to exact calibration and model definitions.
metric_anchor	Record the scale source, value, uncertainty and method.
privacy_mode / raw_media_retained	Make raw-frame retention an explicit governance choice.
started_at / metadata	Order sessions and retain profile-specific context.

5.2 Future frame-observation packet

`FrameObservationPacket` is the planned boundary between Android camera/model code and KC Atlas. It contains camera pose, intrinsics, model hash, keyframe status, dynamic-mask status and a list of typed frame proposals. The packet content is hashed. Converting it to observations still does not commit map state.

```
Android camera / IMU / model
-> FrameObservationPacket
-> Observation records
-> application policy creates SpatialEventProposal
-> verifier / deterministic commit
-> SpatialLedger
```

5.3 Candidate model heads

Head	Proposed output	Ledger treatment
Motion	Relative SE(3) transform and uncertainty.	PoseEstimate; no topology change by itself.
Geometry	Inverse depth, normals, planes/curves.	Observation attributes with error contract.
Association	Stable landmark/scene-node candidate.	Target ID proposal, verified before binding.
Dynamic mask	Static/movable/person/vehicle/unknown.	Dynamic flag and privacy/policy gates.
Topology	Doorway, passage, room boundary, stair or junction.	Route/node proposal, not direct mutation.
Scale	Metric estimate plus calibration status.	Metric anchor and IntervalEstimate.
Confidence	Calibrated covariance/error interval.	Confidence floor and numeric-error guard.

NON-CLAIM

No monocular SLAM network, phone camera pipeline or metric-accuracy benchmark is claimed by 3.9.3. The packet and verifier are executable; the perception front end remains future engineering.

6. Uncertainty, Support and Provenance

6.1 Closed measurement interval

```
estimate = (value, lower, upper, unit, confidence, method, source_ids)
require lower <= value <= upper
```

Interval overlap, not visual similarity alone, drives the repeat-scan measurement classification. Units must match. A measurement without a declared interval can be represented as a zero-width interval, but that does not make it externally accurate.

6.2 Pose uncertainty

PoseEstimate stores translation, normalized quaternion, per-axis translation sigma, per-axis rotation sigma and reference frame. The maximum translation sigma participates in displacement-change classification and application readiness.

6.3 Finite support volumes

Support	Admission rule	Use
none	Admit all points.	Only for deliberately global records.
AABB	Absolute coordinate deltas stay inside half-extents.	Room/component bounds and coarse fixture support.
sphere	Distance from center is at most radius.	Local landmark or dual-sensor support.
cone	Radius plus angular cosine test against axis.	Camera frustum/ray-sector proposal support.
frustum	Finite bounded proxy in the reference implementation.	Future camera-volume refinement.

6.4 Observation provenance

An observation hash covers the capture ID, timestamp, semantic class, target, frame IDs, pose, support, confidence, error, attributes and optional source URI. The loader rejects a mismatched stored provenance hash. The hash supports integrity checks and cache identity; it is not a digital signature, chain of custody or authorship proof.

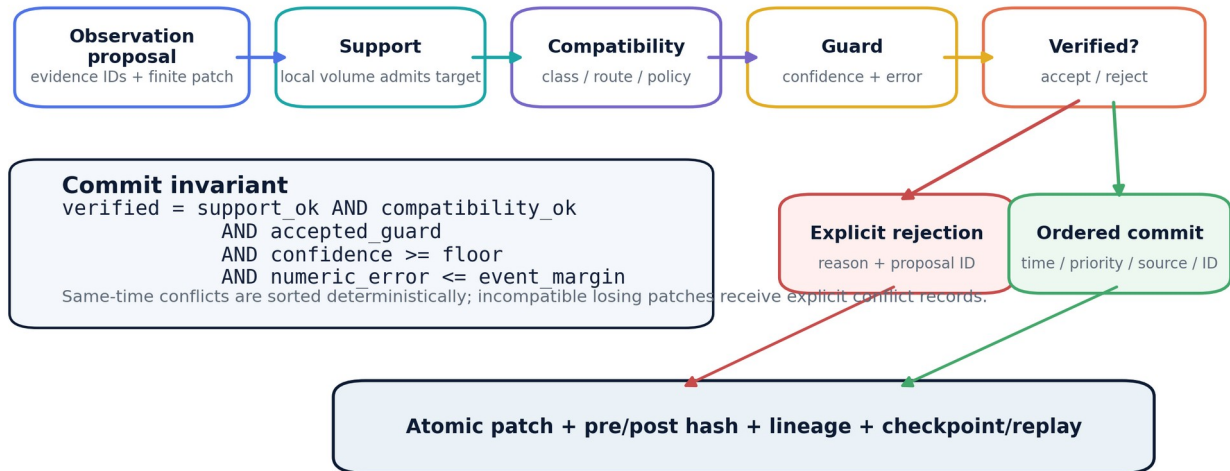
GUARD MARGIN RULE

A packed key, quantized pose, model output or derived mesh is admissible only when its total numerical error remains below the declared event margin and cannot change event ordering.

7. Verified Event Authority, Conflicts and Replay

Verified spatial-event authority

Parallel detection is allowed; unchecked perception output never becomes map authority.



Only the green branch mutates the authoritative ledger. Rejection evidence remains inspectable.

Figure 2. Unchecked detection output is never state authority. Only the verified green branch applies a ledger patch.

7.1 Verification rule

```

verified = support_ok
and compatibility_ok
and guard_status in accepted_statuses
and confidence >= confidence_floor
and numeric_error <= event_margin
  
```

7.2 Deterministic order and conflicts

Verified proposals are sorted by event time, descending priority, source and proposal ID. Same-time incompatible writes to the same field use the first ordered winner; losing proposals receive explicit conflict rejection events. Non-conflicting patches may commit at the same event time.

7.3 Replay

Every committed event carries a sequence number and pre/post state hash. Replay starts from an initial state or checkpoint, reapplies accepted patches and stops at the first divergence. The reference demonstration reaches the same current state hash through replay as through the committed ledger.

8. Map Identity, Route Topology and Queries

8.1 Route-edge contract

Field	Meaning
source / target	Stable node endpoints; edge identity is independent of geometry.
route_class	walk, ramp, stairs or an application-defined class.
status	passable/open, blocked, restricted, unknown, closed or hazard.
allowed_modes	walk, wheelchair, responder or application-defined modes.
clearance	Closed interval; conservative routing uses the lower bound.
slope_degrees	Closed interval; conservative routing uses the upper bound.
confidence	Minimum evidence confidence required by the query/profile.
tags / lineage	Avoidance policy, context and transition history.

8.2 Constraint-aware shortest route

```
RouteQuery = (mode, minimum_clearance, maximum_slope,
              allow_stairs, allow_unknown, confidence_floor, avoid_tags)
```

The deterministic Dijkstra reference solver rejects edges that do not satisfy status, allowed mode, confidence, clearance, slope, stair and tag policy. Unknown is not treated as passable unless the query explicitly allows it. Reachability is also available as a separate query.

8.3 Topology-first value

A sparse room-and-passage graph can already support accessibility review, evacuation planning or incident coordination before a watertight mesh exists. This is a key reason to keep topology, measurements and unknown regions separate from presentation geometry.

9. SafeRoute + DamageDelta Reference Demonstration

SafeRoute topology changes through one verified edge-state event

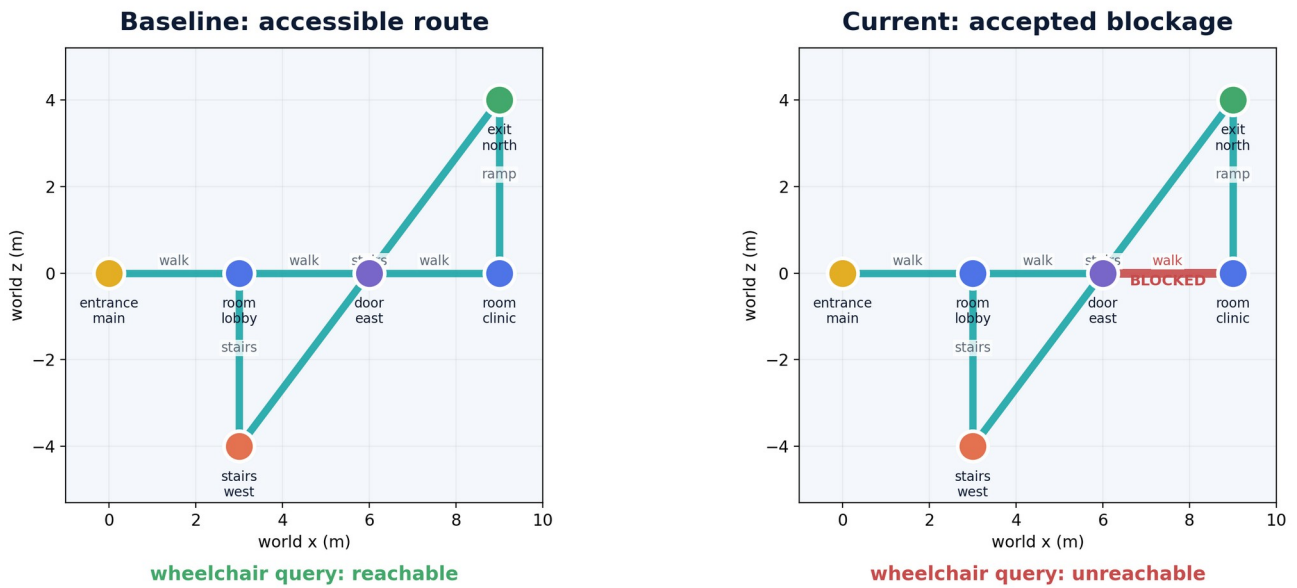


Figure 3. Synthetic route topology before and after one verified blockage event. The alternative stair route remains available to walking/responder modes, not wheelchair mode.

9.1 Captured deterministic results

Metric	Baseline / current result
Ledger nodes / edges	7 stable nodes / 6 route edges.
Observations	7 baseline; 10 repeat-scan observations.
Accepted events	3: blocked passage, beam displacement candidate and doorway remeasurement.
Baseline state hash	bdd4849370b7ce88d607007ff0779c3dceda9b22dc9f45e64d41e9e01d80a049
Current state hash	a49c0f46ba0a0e00b756398b5742134ad45436b54b1448c84ef7517cae6379a6
Replay state hash	a49c0f46ba0a0e00b756398b5742134ad45436b54b1448c84ef7517cae6379a6
Change candidates	16 total: 3 confirmed, 13 within tolerance.
Offline bundle	134,910 bytes; self-contained HTML/JSON/manifest.

INTERPRETATION

The baseline constrained wheelchair query is reachable. After the accepted blockage, the same query is unreachable. This is a deterministic result in synthetic reference data, not a claim that a real corridor, building or evacuation route is safe.

10. Repeat-Scan Change Calculus

Uncertainty-aware repeat-scan classification

Stable IDs are matched first. Closed measurement intervals prevent point-estimate overclaiming.

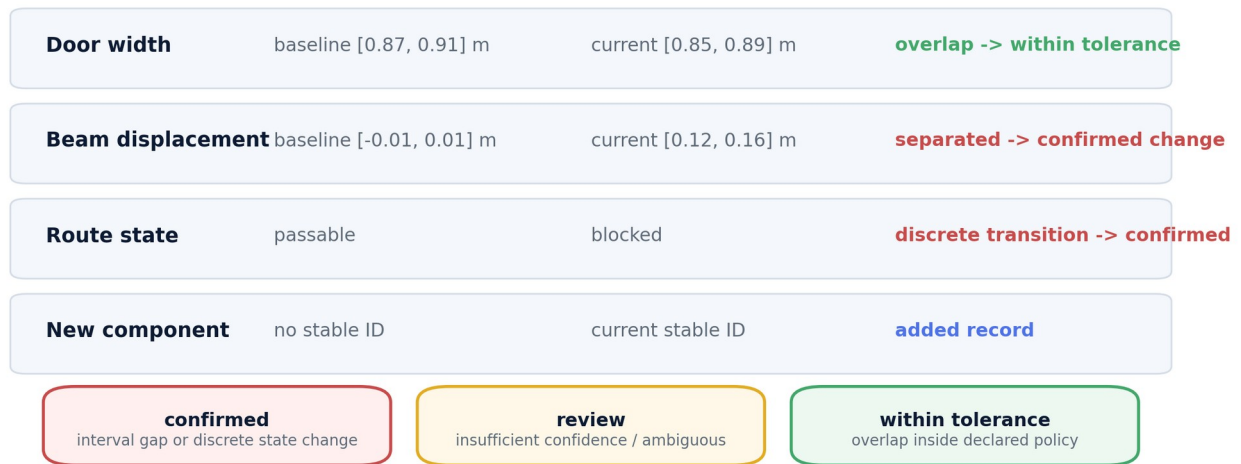


Figure 4. Stable IDs are compared first; interval overlap prevents point-estimate overclaiming.

10.1 Change policy

ChangePolicy declares per-measurement thresholds, position threshold, confidence floor, interval margin and missing-record behavior. The comparison produces canonical candidates and a summary keyed by classification.

Classification	Meaning
confirmed	Separated intervals beyond policy, a discrete state change, or an accepted add/remove under the selected policy.
review	Evidence exists but confidence, uncertainty or policy is insufficient for a definitive classification.
within_tolerance	Intervals overlap or differences remain inside the declared tolerance.
added / removed	Stable identity appears in only one ledger; policy decides whether this is a confirmed change or review.
unchanged	No relevant difference was found.

10.2 Why stable identity matters

Dense point-cloud differencing can confuse viewpoint, meshing, lighting and sampling changes with environmental change. KC Atlas first aligns logical IDs and then compares measurements, pose and route status under explicit uncertainty. Registration quality still remains an application-specific prerequisite.

11. High-Impact Application Profiles

Seven high-impact application profiles

One common evidence substrate; each profile adds its own readiness and claims boundary.

Profile	Primary domain	Authoritative output	Required boundary	Stage
SafeRoute	Accessibility / evacuation	route graph + clearance	not safety certification	MVP
RescueTwin	Disaster response	mergeable event deltas	not structural clearance	next
DamageDelta	Infrastructure	repeat-scan candidates	not engineering diagnosis	MVP
PipeGraph	Utilities	centerline / junction graph	not leak certification	next
FitScan	Assistive / clinical	measurement pre-capture	not diagnosis/fabrication	validation-heavy
HeritageLedger	Cultural heritage	provenance + before/after	hash != legal custody	next
EcoScan	Environment	growth / erosion change	not species/compliance	next

Recommended first product: combined SafeRoute + DamageDelta. It exercises route topology, measurements, repeat scans, replay and offline publication without requiring rescue-grade or clinical certification.

Figure 5. Seven profiles share one evidence substrate but retain distinct readiness, privacy and claims boundaries.

11.1 Profile definition

Each profile is a content-addressed governance record containing purpose, required node classes, required measurements, minimum confidence, optional route query, privacy requirements, safety boundary, forbidden claims and implementation status. Passing readiness means only that the ledger satisfies the machine-readable profile.

RECOMMENDED FIRST BUILD

SafeRoute + DamageDelta is the strongest first product. It exercises route topology, stable identity, uncertainty intervals, repeat scans, deterministic event commit, replay and offline reports without immediately depending on rescue-grade or clinical-grade certification.

11.2 SafeRoute and RescueTwin

SafeRoute

Scan corridors, rooms, doors, ramps, stairs and temporary obstacles into an accessibility/evacuation graph.

Ledger objects: room, door, exit, route edges, doorway width, clearance and slope intervals

Primary output: constrained accessible route and before/after passability report

Boundary: Observed constraints do not certify building-code compliance, fire safety or guaranteed navigation.

SafeRoute minimum viable product

- One explicit metric-scale anchor and calibration record.
- Floor/wall/door/stair/obstacle observations with unknown regions retained.
- Wheelchair/walk/responder route policies.
- Repeat scan that can make a route passable, blocked, restricted or unknown through a verified event.
- Offline report with supporting evidence IDs and measurement intervals.

RescueTwin

Merge route and hazard evidence from multiple phones into an offline operational map.

Ledger objects: room/void/entrance/exit nodes, route edges, hazard observations, checkpoints

Primary output: low-bandwidth verified event deltas and last-known reachability

Boundary: The map cannot certify structural safety, air quality, clearance for entry or responder safety without independent models and sensors.

RescueTwin integration rule

Devices should exchange verified topology and event deltas before dense assets. Conflicting observations remain explicit review items. The designated authority controls commit order and checkpoints.

11.3 DamageDelta and PipeGraph

DamageDelta

Align repeated ledgers and report meaningful component, clearance, opening and route-state changes.

Ledger objects: stable component IDs, measurements, pose uncertainty, baseline/current hashes

Primary output: reviewable change candidates with interval evidence and lineage

Boundary: A candidate is inspection triage, not a structural diagnosis or professional certification.

DamageDelta priorities

- Stable logical component identity across rescans.
- Declared minimum detectable change and registration error.
- Separate geometry, collision and measurement tolerances.
- Change report with confirmed/review/within-tolerance classes.

PipeGraph

Represent visible pipes, conduits, cables and ducts as centerline/radius graphs with junctions.

Ledger objects: pipe_segment, junction, diameter interval, centerline, occlusion/unknown flags

Primary output: compact utility graph and reroute/change history

Boundary: Visible geometry alone cannot diagnose leaks, pressure, material state, electrical safety or capacity.

PipeGraph source fit

The retained swept-tube and parallel-transport-frame mechanisms are a strong match. The authoritative representation should be centerline, radius profile, junction topology and evidence history; a tube mesh remains a derived display asset.

11.4 FitScan and HeritageLedger

FitScan	Capture bounded body or seating geometry for clinician/technician review. Ledger objects: body surface, anatomical landmark, circumference/length intervals, consent/privacy metadata Primary output: pre-capture package and repeat measurement Boundary: Not diagnosis, prescription, fabrication authority or guaranteed fit. Dedicated calibration and clinical validation are mandatory.
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FitScan privacy gate

Derived-only publication should be the default. Raw media, personal identifiers and precise location require explicit consent and access policy. Profile readiness cannot replace medical-device or health-data obligations.

HeritageLedger	Document sites and components after damage, conflict, fire, flood or vandalism. Ledger objects: site/component IDs, capture/model hashes, geometry evidence, before/after events Primary output: integrity-checkable offline evidence bundle and chronology Boundary: Hashes support integrity checking but do not by themselves prove authenticity, ownership or legal chain of custody.
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HeritageLedger value

Content-addressed definitions and replay preserve which calibration/model/profile created each result. Sensitive location precision may be reduced in public views while the restricted ledger retains the authoritative coordinates.

11.5 EcoScan and Product Sequencing

EcoScan	Repeat spatial records of plants, trunks, canopy, riverbanks or restoration plots. Ledger objects: plot/plant IDs, diameter intervals, counts, change candidates Primary output: growth, loss, erosion or vegetation-change report Boundary: Geometry does not certify species, disease, yield, ecosystem condition or regulatory compliance.
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Lower-risk validation environment

EcoScan offers a practical environment for validating repeat-scan identity, compact indexing and change classification before applying the same infrastructure to safety-critical rescue or clinical workflows.

Recommended sequencing

Stage	Profiles	Reason
1 - MVP	SafeRoute + DamageDelta	Exercises the most substrate mechanisms with bounded claims.
2 - field expansion	EcoScan + PipeGraph + HeritageLedger	Adds domain diversity, offline work and repeated capture.
3 - operational research	RescueTwin	Requires multi-device authority, incident procedure and stronger reliability.
4 - validation-heavy	FitScan	Requires calibration, privacy, clinician workflow and external clinical validation.

12. Finite SCLP-Derived Observation Indexing

Finite 64-bit local observation key

SCLP-derived indexing for local rays/cells; never a replacement for 3D pose, uncertainty, semantics or lineage.

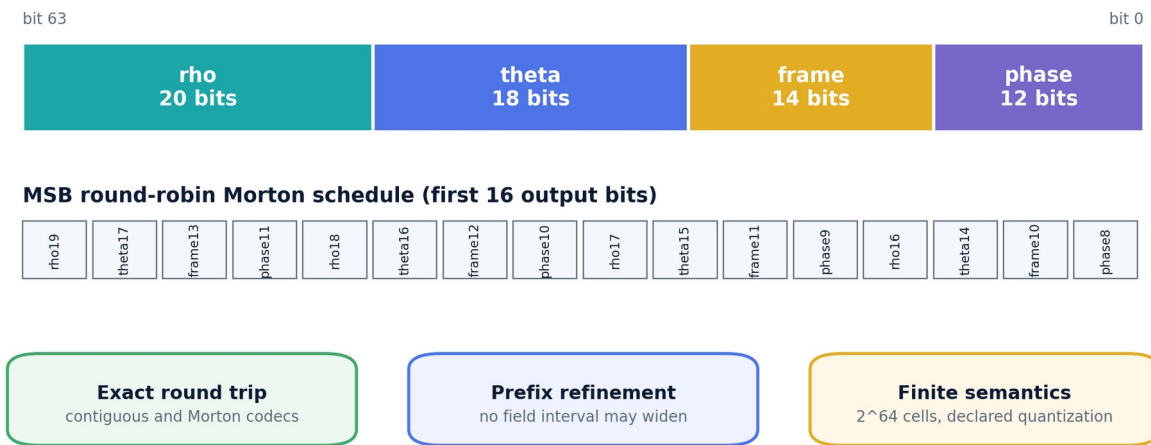


Figure 6. A finite local observation key supports compact indexing and prefix refinement; it does not encode complete map state.

12.1 Field layout

Field	Bits	Meaning
rho	20	Quantized local log radius in a declared finite range.
theta	18	Quantized local angular coordinate.
frame	14	Modular frame/keyframe coordinate.
phase	12	Application-declared phase or local state coordinate.

Two exact codecs are implemented: contiguous ranges and MSB round-robin Morton interleaving. Both round-trip the same quantized tuple and intentionally yield different 64-bit words. Prefix bounds never widen and at least one scheduled field narrows at each appended bit.

12.2 Explicit limitation

FINITE-KEY BOUNDARY

The 64-bit key is a local indexing aid. It does not contain 3D pose, covariance, semantic class, stable identity, route state, source frames or lineage. Key collisions, quantization error and sparse-index overhead must be measured for the target application.

13. Offline Publication, Privacy and Evidence Exchange

13.1 Evidence bundle

Artifact	Contents	Authority
ledger.json	Publication-policy view of metadata, captures, observations, nodes, edges and events.	Authoritative only under the application policy and retained full ledger.
change-report.json	Canonical repeat-scan candidates and summary.	Derived from named baseline/current hashes.
index.html	Inline CSS/SVG/JSON/JavaScript viewer; no network assets.	Presentation and inspection.
manifest.json	Path, byte count and SHA-256 for every output file.	Integrity check; not legal authentication.

13.2 Publication policy

- Remove raw source URIs.
- Pseudonymize author/operator labels.
- Round coordinates to a declared precision.
- Exclude raw media and sensitive metadata.
- Retain evidence counts, state hash, profile boundaries and uncertainty.

13.3 Offline result

The checked-in reference bundle contains four files and 134,910 bytes. Validation found no external HTTP/HTTPS markers, script source references, stylesheet links or import URLs. Its manifest verifies after regeneration.

SECURITY BOUNDARY

Browser storage, hashes and redaction are not a complete security system. Encryption, access control, secure deletion, key management, audit policy and legal data-governance obligations remain deployment responsibilities.

14. Android Spatial-Ledger Bridge Decision

14.1 Executive decision

POSTPONE NEW APP

The phone-native camera/IMU/model implementation is postponed until the 3.9.3 ledger, packet, uncertainty and commit contracts are complete. This prevents a fast prototype from silently turning model output into map authority.

14.2 Retained baseline

Baseline fact	Recorded value
Project	android/UGTSKCKKijTGrove
Retained files	35
Supplied APK	android/UGTSKCKKijTGrove/app/build/outputs/apk/pocoX7Pro/release/app-pocoX7Pro-release.apk
APK bytes	1,040,086
APK SHA-256	79d0e93b10ba57e33a49e8c1338009a3b4fa92f952c864d609bbf8ffcb816cab
3.9.3 status	Preserved; not modified, rebuilt, installed or device-profiled.

14.3 Required future milestones

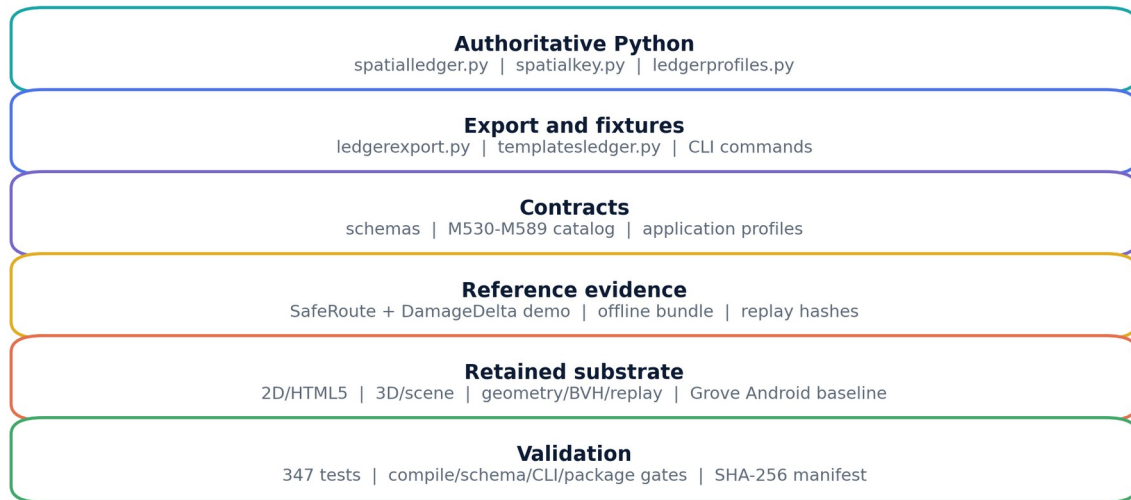
1. Load prerecorded FrameObservationPacket fixtures and reproduce Python-oracle accept/reject decisions.
2. Integrate CameraX/NDK camera capture and optional IMU with exact timestamps and calibration profile.
3. Run a two-rate perception stack: light tracking each frame, heavier model on keyframes/uncertainty spikes.
4. Keep model output proposal-only; commit through the ledger verifier.
5. Export an offline bundle on device and verify it independently.
6. Benchmark a named physical phone with p50/p95/p99 latency, memory, thermal and power results at equal error.

14.4 Phone-performance boundary

The retained POCO profile and supplied APK are useful implementation evidence, but they do not establish capture-model throughput or accuracy. The future benchmark must state the exact phone build, sensor stream, model hash, delegate/backend, sustained duration, calibration protocol and false/missed-event oracle.

15. Reference Implementation and Package Map

3.9.3 package map



The source PDFs are registered by hash; the attached 3.9.2 implementation baseline is retained inside the package.

Figure 7. KC Atlas is additive: the new ledger modules sit beside the retained game, scene and Android baseline.

15.1 New modules

Module	Primary responsibility
spatialledger.py	Records, canonical hashes, validation, proposal commit, replay, routes, change and publication view.
spatialkey.py	Finite log-polar quantization, contiguous/Morton key codecs and prefix bounds.
ledgerprofiles.py	Seven application definitions and readiness validation.
ledgerexport.py	Self-contained HTML report, evidence bundle and manifest verification.
templatesledger.py	Synthetic SafeRoute + DamageDelta ledgers and deterministic demonstration.

15.2 Machine-readable assets

- Spatial-ledger JSON Schema.
- Frame-observation packet JSON Schema.
- M530-M589 CSV/JSON catalog and combined M198-M589 catalog.
- Application-profile registry with definition hashes.
- Source register with SHA-256 values.
- Reference baseline/current/change/bundle files.

16. Command-Line Workflow

Command	Purpose
info	Print release, retained runtime paths and Android postponement.
new-ledger	Create a SafeRoute reference ledger.
validate-ledger	Validate the ledger and optional application profile.
route-ledger	Run a constrained route query.
diff-ledger	Create an uncertainty-aware baseline/current change report.
build-ledger-report	Build a self-contained evidence bundle.
demo-ledger	Regenerate the SafeRoute + DamageDelta vertical slice.

```
PYTHONPATH=src python -m ugts_kc3 demo-ledger examples/kc_atlas_spatial_ledger
```

```
PYTHONPATH=src python -m ugts_kc3 validate-ledger \
  examples/kc_atlas_spatial_ledger/baseline-ledger.json \
  --profile safe-route
```

```
PYTHONPATH=src python -m ugts_kc3 route-ledger \
  examples/kc_atlas_spatial_ledger/baseline-ledger.json \
  entrance-main exit-north --mode wheelchair \
  --minimum-clearance 0.80 --maximum-slope 8 --no-stairs --json
```

```
PYTHONPATH=src python -m ugts_kc3 diff-ledger \
  baseline-ledger.json current-ledger.json --output change-report.json
```

16.1 Python distributions

The 3.9.3 wheel contains 74 archive entries and is 186,916 bytes. The source distribution contains 101 entries and is 185,388 bytes. A clean target-directory install imports the new ledger types and reports version 3.9.3.

17. Validation and Release Evidence

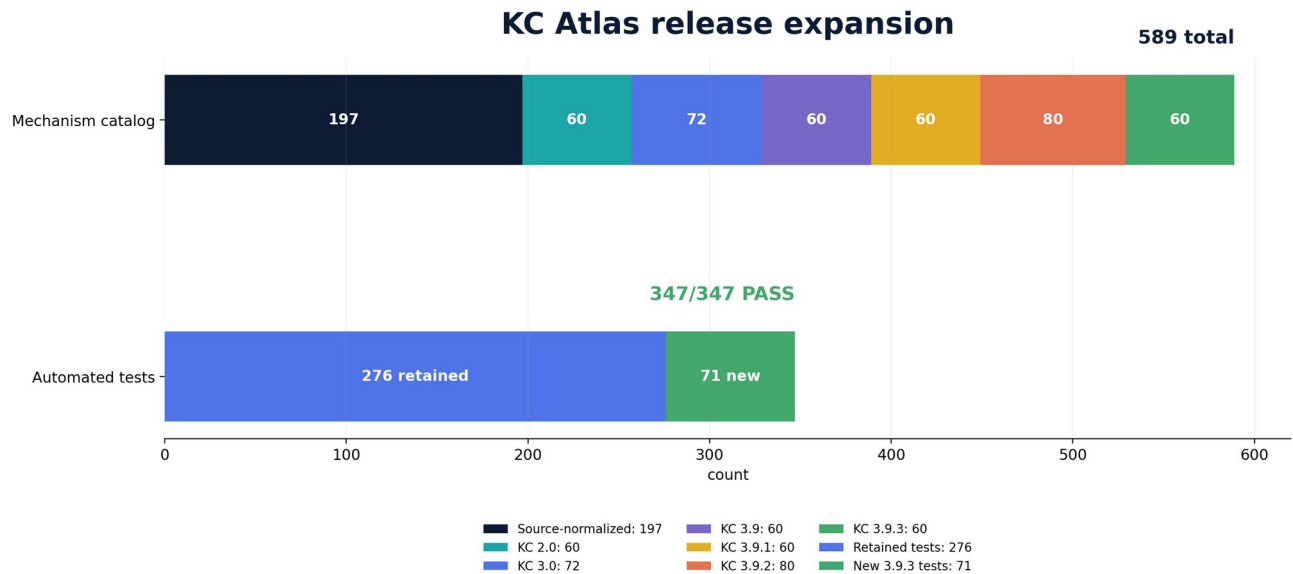


Figure 8. 3.9.3 adds 60 mechanisms and 71 tests while retaining the earlier implementation layers.

Gate	Captured result
Python compile	PASS - all source modules compile with compileall.
Automated tests	347/347 PASS; 276 retained + 71 new.
Schemas	Baseline, pre-event, current and frame-packet examples validate.
Catalog	M198-M589 contiguous; M530-M589 count is 60.
Demo replay	Current state hash equals replay state hash.
Route regression	Baseline accessible route succeeds; blocked current route returns unreachable.
Offline bundle	Manifest verifies and no external network assets are referenced.
Profiles	Seven definition hashes and registry hash verify.
Source basis	Seven registered source hashes match.
Distributions	Wheel/sdist contain new modules; wheel clean-install passes.
Android	Retained project/APK recorded; no new build/device run claimed.

17.1 Validation interpretation

These gates establish internal consistency, reproducibility and bounded behavior of the included reference implementation. They do not establish real-world monocular scale accuracy, route safety, structural condition, clinical fitness, legal evidence status or a performance advantage over conventional mapping systems.

18. Mechanism Catalog M530-M589

KC Atlas adds 60 engineering mechanisms. Full definitions, formulas, integration notes and validation statuses are supplied in CSV and JSON under `spec/`.

Range	Domain	Count	Representative mechanisms
M530-M530	Spatial ledger core	1	Versioned spatial-ledger schema
M531-M534	Capture and provenance	4	CaptureProfile, Metric anchor contract, Calibration and model content hashes...
M535-M537	Perception adapter	3	Frame observation packet, Typed frame proposal, Dynamic-mask declaration
M538-M545	Uncertainty and support	8	Pose uncertainty record, Closed scalar interval estimate, Finite support volume...
M546-M553	Map authority	8	Stable map-node identity, Stable route-edge identity, Evidence/state separation...
M554-M561	Topology and navigation	8	Route graph query, Explicit route class, Clearance uncertainty interval...
M562-M569	Repeat-scan change	8	Repeat-scan comparison contract, Stable-ID node and edge diff, Interval-aware measurement difference...
M570-M577	Application profiles	8	SafeRoute profile, RescueTwin profile, DamageDelta profile...
M578-M583	Publication and exchange	6	PublicationPolicy, Precision-bounded publication view, Raw source URI redaction...
M584-M587	Finite indexing	4	SCLP-derived local observation key, Contiguous 64-bit observation layout, Morton round-robin observation layout...
M588-M589	Android bridge boundary	2	Spatial-ledger Android adapter contract, Retained Grove baseline and physical-device gate

18.1 Catalog continuity correction

M529 CLOSURE

The supplied 3.9.2 catalog declared M450-M529 but its machine-readable list ended at M528. Version 3.9.3 adds one explicit retained-baseline catalog-closure row at M529, marked `corrected_in_3.9.3`, before beginning the new M530 namespace. No unobserved Android function was invented.

18.2 Catalog meaning

Mechanism IDs are engineering traceability entries. They are not by themselves claims of novelty, patentability, ownership, independent scientific validation or product certification.

Appendix A - Compact M530-M589 Index

The table below is a compact human-readable index. The machine-readable catalog remains authoritative for full definitions and formulas.

ID	Domain	Mechanism	Substrate integration	Validation
M530	Spatial ledger core	Versioned spatial-ledger schema	Creates one source of truth for high-impact spatial applications.	Implemented + schema + tests
M531	Capture and provenance	CaptureProfile	Makes acquisition assumptions explicit and replayable.	Implemented + tested
M532	Capture and provenance	Metric anchor contract	Prevents uncalibrated image geometry from masquerading as metric measurement.	Implemented record + validated
M533	Capture and provenance	Calibration and model content hashes	Supports reproducibility and model-version audit without treating a hash as ownership proof.	Implemented record + tested
M534	Capture and provenance	Capture privacy mode	Supports field, health and heritage privacy boundaries.	Implemented + tested
M535	Perception adapter	Frame observation packet	Separates perception output from authoritative map mutation.	Implemented + schema + tests
M536	Perception adapter	Typed frame proposal	Allows heterogeneous models to share one verifier boundary.	Implemented + tested
M537	Perception adapter	Dynamic-mask declaration	Reduces person/vehicle contamination in persistent maps.	Implemented + tested
M538	Uncertainty and support	Pose uncertainty record	Enables uncertainty-aware change classification and display.	Implemented + tested
M539	Uncertainty and support	Closed scalar interval estimate	Prevents point estimates from losing measurement uncertainty.	Implemented + tested
M540	Uncertainty and support	Finite support volume	Retains the canonical local-support stage.	Implemented + tested
M541	Uncertainty and support	Observation provenance hash	Detects accidental evidence mutation.	Implemented + tested
M542	Uncertainty and support	Multi-keyframe evidence set	Supports review and reprocessing.	Implemented + tested
M543	Uncertainty and support	Application confidence floor	Makes acceptance policy explicit.	Implemented + tested
M544	Uncertainty and support	Numeric error versus event margin	Preserves packed/numerical boundary discipline.	Implemented + tested
M545	Uncertainty and support	Explicit unknown state	Prevents unsafe completion of unobserved space.	Implemented + tested
M546	Map authority	Stable map-node identity	Supports repeat scans and lineage.	Implemented + tested
M547	Map authority	Stable route-edge identity	Makes passable-to-blocked change a typed event.	Implemented + tested
M548	Map authority	Evidence/state separation	Avoids direct model writes to authority.	Implemented + tested
M549	Map authority	Canonical ledger state hash	Provides deterministic identity across JSON round trips.	Implemented + tested
M550	Map authority	Spatial event proposal	Provides one commit record across applications.	Implemented + tested
M551	Map authority	Support-compatibility-guard verification	Retains the canonical UGTS authority chain.	Implemented + tested

ID	Domain	Mechanism	Substrate integration	Validation
M552	Map authority	Deterministic proposal ordering and conflict rejection	Makes commits and conflict outcomes replayable.	Implemented + tested
M553	Map authority	Checkpoint and hash-verified replay	Supports audit and deterministic reconstruction.	Implemented + tested
M554	Topology and navigation	Route graph query	Supports accessibility, rescue and maintenance routing.	Implemented + tested
M555	Topology and navigation	Explicit route class	Prevents overlapping paths from collapsing identity.	Implemented + tested
M556	Topology and navigation	Clearance uncertainty interval	Enables conservative accessibility filtering.	Implemented + tested
M557	Topology and navigation	Slope uncertainty interval	Supports ramp/steepness constraints without false precision.	Implemented + tested
M558	Topology and navigation	Constraint-aware shortest route	Implements SafeRoute query semantics.	Implemented + tested
M559	Topology and navigation	Reachability query	Supports evacuation/rescue region analysis.	Implemented + tested
M560	Topology and navigation	Passable/blocked/restricted/unknown status	Makes operational topology explicit.	Implemented + tested
M561	Topology and navigation	Route-transition lineage	Supports before/after route evidence.	Implemented + tested
M562	Repeat-scan change	Repeat-scan comparison contract	Separates registration from change classification.	Specified + demonstrated
M563	Repeat-scan change	Stable-ID node and edge diff	Prevents nearest-coordinate identity substitution.	Implemented + tested
M564	Repeat-scan change	Interval-aware measurement difference	Reduces false certainty in damage and growth reports.	Implemented + tested
M565	Repeat-scan change	Pose displacement with uncertainty	Distinguishes drift from plausible movement.	Implemented + tested
M566	Repeat-scan change	Added/removed identity candidate	Supports inventory and heritage documentation.	Implemented + tested
M567	Repeat-scan change	Route-status change candidate	Supports SafeRoute and RescueTwin deltas.	Implemented + tested
M568	Repeat-scan change	Change classification ledger	Separates machine triage from professional judgment.	Implemented + tested
M569	Repeat-scan change	Canonical change report	Makes threshold choices and evidence auditable.	Implemented + schema + tests
M570	Application profiles	SafeRoute profile	Highest-feasibility first product.	Implemented + demonstrated
M571	Application profiles	RescueTwin profile	High societal impact with explicit safety boundary.	Implemented profile; field validation required
M572	Application profiles	DamageDelta profile	Supports inspection candidates without engineering certification claims.	Implemented + demonstrated
M573	Application profiles	PipeGraph profile	Uses retained sweep and stable graph identity.	Implemented profile
M574	Application profiles	FitScan profile	Supports communication, not diagnosis or fabrication authority.	Implemented profile; clinical validation required
M575	Application profiles	HeritageLedger profile	Supports integrity checking without legal ownership claims.	Implemented profile
M576	Application profiles	EcoScan profile	Provides a lower-risk validation domain.	Implemented profile
M577	Application profiles	Profile readiness validator	Prevents a generic ledger from silently claiming application readiness.	Implemented + tested

ID	Domain	Mechanism	Substrate integration	Validation
M578	Publication and exchange	PublicationPolicy	Separates authoritative storage from shareable view.	Implemented + tested
M579	Publication and exchange	Precision-bounded publication view	Reduces unnecessary location precision and stabilizes reports.	Implemented + tested
M580	Publication and exchange	Raw source URI redaction	Supports derived-only workflows.	Implemented + tested
M581	Publication and exchange	Self-contained offline HTML report	Extends the 3.9 offline deployment idea to spatial evidence.	Implemented + tested
M582	Publication and exchange	Evidence-bundle manifest	Supports transfer integrity and reproducibility.	Implemented + tested
M583	Publication and exchange	Evidence-bundle verification	Detects corruption after offline transfer.	Implemented + tested
M584	Finite indexing	SCLP-derived local observation key	Reuses SCLP packing with an explicit application boundary.	Implemented + tested
M585	Finite indexing	Contiguous 64-bit observation layout	Supports deterministic keys and exact round trips.	Implemented + tested
M586	Finite indexing	Morton round-robin observation layout	Supports prefix grouping while remaining distinct from contiguous layout.	Implemented + tested
M587	Finite indexing	Prefix refinement bounds	Supports bounded candidate refinement.	Implemented + tested
M588	Android bridge boundary	Spatial-ledger Android adapter contract	Records the implementation path while deliberately postponing the native app.	Specified; app postponed
M589	Android bridge boundary	Retained Grove baseline and physical-device gate	Bases later work on the working implementation without overstating 3.9.3 delivery.	Baseline preserved; no new APK claimed

19. Migration, Kill Criteria and Final Definition

19.1 Migration from 3.9.2

1. Retain the original game/mobile project and Android files as source artifacts.
2. Create a CaptureProfile with coordinate, scale, calibration, model and privacy metadata.
3. Derive hash-verifiable observations with finite support and uncertainty.
4. Create stable map nodes and route edges; do not use coordinates as identity.
5. Commit map changes through SpatialEventProposal and checkpoint/replay.
6. Publish a reduced evidence view only under an explicit PublicationPolicy.

19.2 Numerical and authority kill criteria

- Pose, quantization, model or registration error exceeds the event margin or changes event order.
- Unknown/low-confidence observations are silently promoted to passable or confirmed states.
- Support and compatibility fail to prune enough candidates to justify the added substrate cost.
- Same-time conflicts cannot be resolved without hidden application logic.
- Replay diverges from committed state or source/model/calibration hashes cannot be verified.
- Change classification is dominated by registration/lighting/mesh drift rather than real change.
- A conventional map/database is simpler, faster or more accurate at equal error.

19.3 Application kill criteria

- Accessibility output is presented as a legal or safety certification.
- Disaster mapping is used to clear structural entry without an independent structural assessment.
- Damage candidates are presented as diagnoses.
- Visible utility geometry is used to infer leak/pressure/electrical condition without supporting sensors.
- FitScan output is used for diagnosis, prescription or fabrication without clinical validation.
- Hashes are presented as proof of ownership, authenticity or complete legal chain of custody.
- EcoScan geometry is presented as species, disease, yield or compliance certification.

19.4 Final definition

FORMAL DEFINITION

UGTS-KC 3.9.3 KC Atlas is a finite, content-addressed spatial evidence and change ledger in which captures and observations remain proposal evidence; local support, semantic compatibility, confidence and numerical-error guards determine verified events; deterministic commits update stable map and route identity; checkpoints and hashes enable replay; interval-aware comparisons produce change candidates; application profiles and publication policies bound downstream use; and Android perception remains a future adapter rather than map authority.

Appendix B - Package, Source and Attribution Register

B.1 Final package map

Path	Contents
src/ugts_kc3/	Retained runtime plus new spatial ledger, key, profile, export and template modules.
examples/kc_atlas_spatial_ledger/	Baseline/current ledgers, frame packet, change report, replay summary and offline bundle.
spec/	Schemas, contracts, source register and catalogs through M589.
docs/	Release, guide, migration, Android postponement and build status.
report/	This PDF/DOCX and retained legacy reports.
tests/	347-test retained-plus-new suite.
validation/	Compile, tests, schema, demo, CLI, distribution, source, Android and manifest evidence.
android/UGTSKCKKijTGrove/	Retained requester-supplied 3.9.2 Android implementation baseline and APK.

B.2 Source hashes

Registered source file	SHA-256
UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	d7ee3c91abf60e35b4b15fb296477974e45802c24e1fde618d683949bc1c7df2
UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	a29dd9959ebd300c32d0e1df8fa2c0d4ee15f428892557b98b5f18cd3e878399
UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	183ffa4e2bf29cfa0308687cd221529321598dbd69ef75c150abc0c64810f214
Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	3a6b6c57118ca4e2d7d7e577508c3477a095d133791f3708cb56a054285225d2
UGTS_KC_3_6_Tom_Klootwijk.pdf	e21433cd8378368d4e4ae278d3c9c364725e45ae7fd75b743328bd2d6003a46
UGTS_KC_K_Kij_T_Grove_3_9_2_COMPLETE_ALL_FILES - Tom Klootwijk(1).zip	2d0197bb4526ca994eb67f68985ebd349fb91f52f562870b97f01d03023f8944

B.3 Attribution notice

Prepared for Tom Klootwijk as the UGTS-KC 3.9.3 KC Atlas Pocket Spatial Ledger edition. The earlier package preserves requester-supplied identity, date and KC attribution records. Those assertions were not independently verified and are not repeated here as evidence of authorship, ownership, priority, licensing status or chain of title.

B.4 Release identifiers

Identifier	Value
Runtime version	3.9.3
Codename	KC Atlas
Spatial-ledger schema	ugts-kc-spatial-ledger-3.9.3
Frame packet schema	ugts-kc-frame-observation-packet-3.9.3
Mechanism namespace	M001-M589 by combined source/catalog
Automated tests	347/347 PASS
Android capture app	Postponed; 3.9.2 Grove baseline retained
Report source	report/ UGTS_KC_3_9_3_KC_ATLAS_POCKET_SPATIAL_LEDGER.docx